OPENCV AI COMPETITION 2026 · AGENTIC VISION

Northbridge Watch

Quiet-by-default home vision. Classical OpenCV 5 sees the porch and kitchen, and a small policy agent decides: stay silent, log, notify one person, or ask a human.

Team Northbridge · household of Avery & Morgan (pseudonyms)

OpenCV 5.0.0 · Python · Vercel (live) · AWS Lambda · S3 · SNS — deploy-ready, not yet deployed

THE PROBLEM

Home cameras spam alerts. Nobody wants a ping for an empty porch.

After a week of “motion at front door”, people mute the camera — and then miss the one alert that mattered: a package left in the rain, a spill someone could slip on, an empty fridge before the weekend.

Avery & Morgan want the opposite: a camera that stays quiet unless something really changed, and that asks a person when the stakes are real.

OUR SOLUTION

Quiet by default. Perception changes the plan.

Every frame ends in exactly one of four actions, with a full trace: perception → decision → speakable message → action JSON.

`noop`

Calm porch or kitchen. Nobody is pinged.

`auto_log`

Low-stakes event, logged quietly (and during quiet hours).

`notify`

One person, one short message: “A package arrived on the porch.” Empty fridge → restock note for Morgan.

`human_required`

Safety events (floor spill) always go to a human; Avery & Morgan must acknowledge.

Scenes: calm porch · package at door · busy porch (2+ packages, doorway blocked → escalate) · kitchen floor spill · empty vs. full fridge shelf.

HOW IT WORKS

Classical OpenCV 5 — no black box

`cv.resize`

Fixed 640×480 working frame (fixes half-res failures)

→

`cvtColor HSV · inRange · morphologyEx`

Cardboard and liquid masks

→

`findContours · approxPolyDP · convexHull`

Box vs. spill by shape and solidity

→

`Sobel · Canny · occupancy gate`

Shelf lines, floor tiles, empty vs. full fridge (4% gate)

Spill = safety hazard → **always a human**

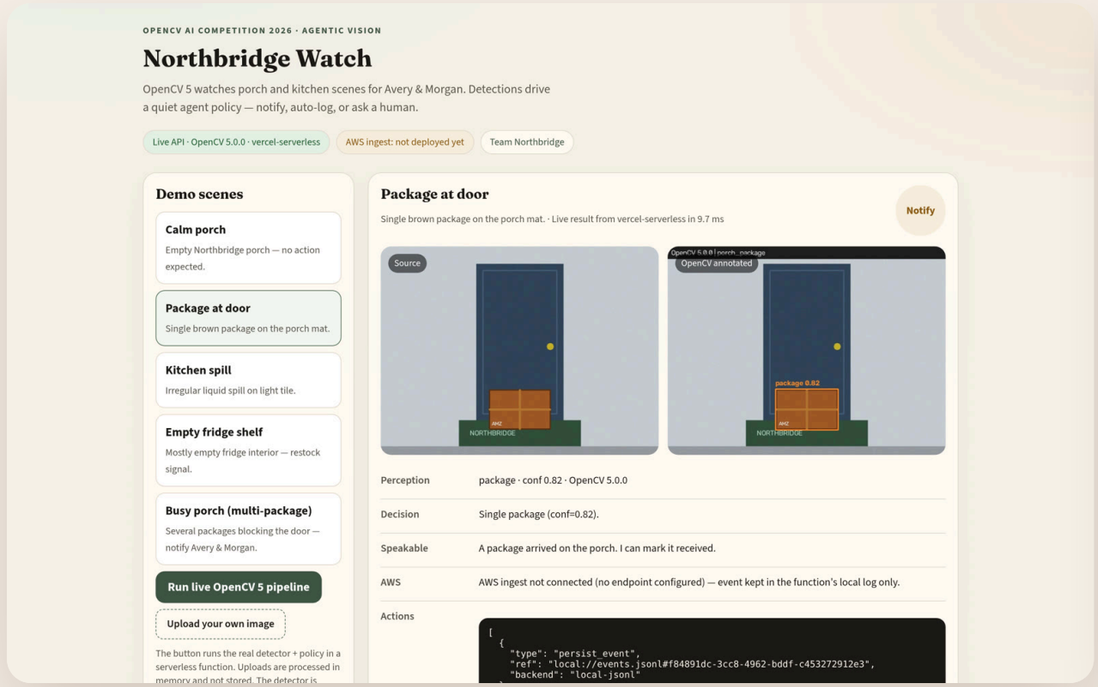
≥ 2 packages → doorway blocked → escalate

Calm → **never** notifies · quiet hours hold non-urgent pings

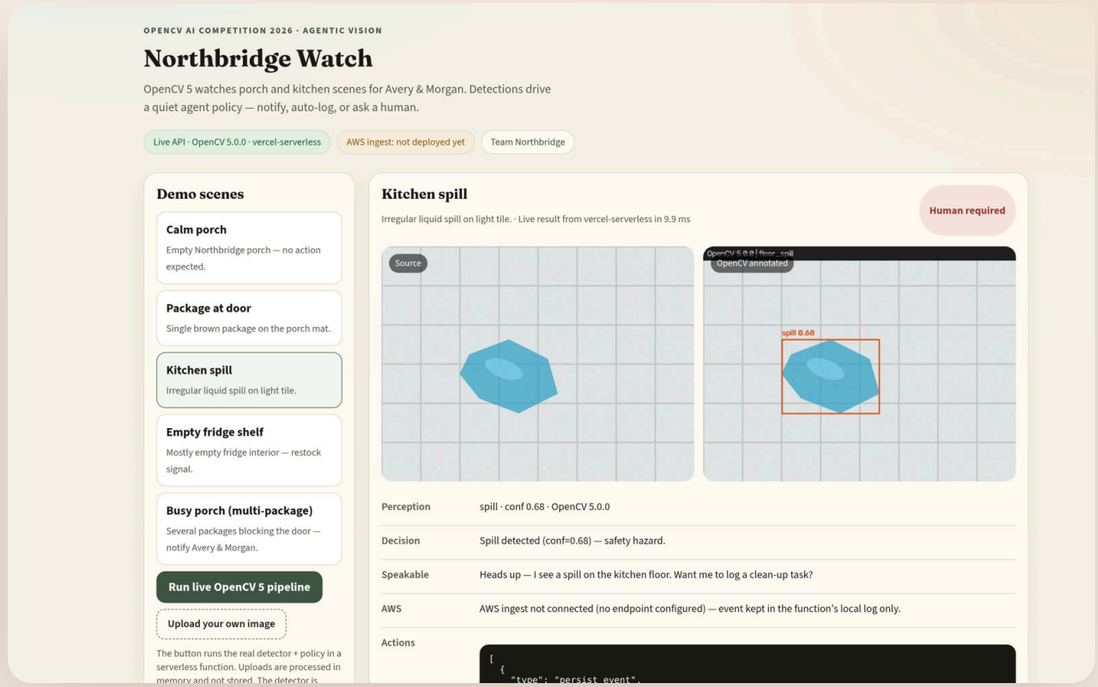
Deterministic `PolicyEngine` + `AgentOrchestrator` in one Python package, 26 unit tests, one-command demo (`make demo`).

LIVE DEMO · NORTHBRIDGE-WATCH.VERCEL.APP

The real pipeline, running serverless



Package at door → **notify** Avery.



Kitchen spill → **human_required**, Avery & Morgan.

Measured on the live URL (2026-10-25, 15 warm requests): server-side pipeline ≈ 10 ms per frame (9.8–13.7 ms), browser round trip 67–124 ms. Uploads are processed in memory and never stored.

MEASURED RESULTS · HELD-OUT SYNTHETIC TEST SPLIT

Label accuracy, v1 → v2

CONDITION (280 IMAGES EACH)	V1 (ORIGINAL)	V2 (CURRENT)	V2 SPILL RECALL
clean	82.9%	**96.8%**	87.5%
photometric shift	80.7%	**92.1%**	70.0%
blur + noise	83.9%	**97.5%**	87.5%
JPEG q30	83.2%	**96.8%**	87.5%
half resolution (320×240)	44.6%	**96.8%**	87.5%
out-of-domain, 120 images (coffee spill, white mailer, night package)	—	**33.3%**	**0.0%**

7 randomized scene families × 40 per condition; thresholds tuned only on dev seeds, numbers on held-out test seeds. OpenCV 5.0.0. Reproduce: `make eval` / `make baseline`. Synthetic data only — a sanity check, not real-world accuracy. Source: technical report §4.

DEPLOYMENT, LIMITATIONS, RESPONSIBLE USE

Honest about the edges

— **Vercel** LIVE

Same Python + OpenCV 5 package as a serverless function: `/api/analyze`, `/api/health`.

— **AWS Lambda · S3 · SNS**

DEPLOY-READY, NOT YET DEPLOYED

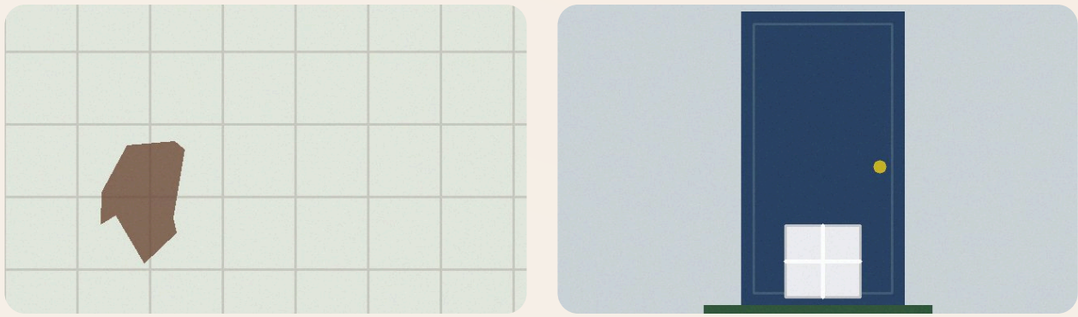
SAM stack (API Gateway → Lambda arm64 → S3 event log + SNS) written and unit-tested; `make deploy-aws` deploys it and wires it into the app.

— **Responsible use:** no faces or people detected or identified; uploads not stored; safety events always go to a human; pseudonymous household.

— Synthetic scenes from our own generator, not real photos — real photos often come back “calm”.

— Colour-dependent: coffee-coloured spills and white mailers are missed.

— Spill recall drops to 70% under lighting changes.



Known misses: coffee spill (left), white mailer (right).

TRY IT

Northbridge Watch

LIVE APP	northbridge-watch.vercel.app
REPORT	github.com/dl88jing/northbridge-watch/blob/main/docs/technical-report.pdf
CODE	github.com/dl88jing/northbridge-watch (Apache-2.0)

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